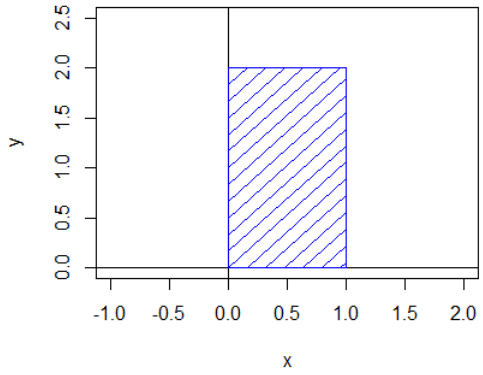
F78PA Probability & Statistics A

Tutorial 8 – Solutions

1. Joint probability density function of X and Y is

$$f_{X,Y}(x,y) = \begin{cases} \frac{6}{7}\left(x^2 + \frac{xy}{2}\right) & 0 \le x \le 1,\ 0 \le y \le 2, \\ 0 & \text{otherwise.} \end{cases}$$

(a) Region of non-zero density:



(b) To verify that $f_{X,Y}(x,y)$ is a valid pdf, need to check that it is non-negative and integrates to 1.
First note that for $0 \le x \le 1$, $0 \le y \le 2$ we have $f_{X,Y}(x,y) = \frac{6}{7}\left(x^2 + \frac{xy}{2}\right) \ge 0$, and $f_{X,Y}(x,y) = 0$ elsewhere, hence $f_{X,Y}(x,y) \ge 0$ for all $(x,y) \in \mathbb{R}^2$.
Next,

$$\begin{aligned} \iint_{\mathbb{R}^2} f_{X,Y}(x,y)\,dy\,dx &= \int_{x=0}^{1}\int_{y=0}^{2} \frac{6}{7}\left(x^2 + \frac{xy}{2}\right) dy\,dx \\ &= \int_{x=0}^{1} \left[\frac{6}{7}\left(x^2 y + \frac{xy^2}{4}\right)\right]_{y=0}^{2} dx \\ &= \int_{x=0}^{1} \frac{6}{7}\left(2x^2 + x\right) dx \\ &= \left[\frac{4x^3}{7} + \frac{3x^2}{7}\right]_{x=0}^{1} \\ &= \frac{4}{7} + \frac{3}{7} = 1 \end{aligned}$$

So $f_{X,Y}(x,y)$ is a valid density function.

(c) Looking at the region of non-zero density, see that $\mathrm{Im}X = [0,1]$. So $f_X(x) = 0$ for $x < 0$ or $x > 1$. For $0 \le x \le 1$,

$$\begin{aligned} f_X(x) &= \int_{y=0}^{2} \frac{6}{7}\left(x^2 + \frac{xy}{2}\right) dy \\ &= \left[\frac{6}{7}\left(x^2 y + \frac{xy^2}{4}\right)\right]_{y=0}^{2} \\ &= \frac{6}{7}\left(2x^2 + x\right) \end{aligned}$$

So marginal density for X is

$$f_X(x) = \begin{cases} \frac{6}{7}\left(2x^2 + x\right) & 0 \le x \le 1, \\ 0 & \text{otherwise.} \end{cases}$$

Looking at the region of non-zero density, see that $\mathrm{Im}Y = [0,2]$. So $f_Y(y) = 0$ for $y < 0$ or $y > 2$. For $0 \le y \le 2$,

$$\begin{aligned} f_Y(y) &= \int_{x=0}^{1} \frac{6}{7}\left(x^2 + \frac{xy}{2}\right) dx \\ &= \left[\frac{6}{7}\left(\frac{x^3}{3} + \frac{x^2 y}{4}\right)\right]_{x=0}^{1} \\ &= \frac{6}{7}\left(\frac{1}{3} + \frac{y}{4}\right) = \frac{2}{7} + \frac{3y}{14} \end{aligned}$$

So marginal density for Y is

$$f_Y(y) = \begin{cases} \frac{2}{7} + \frac{3y}{14} & 0 \le y \le 2, \\ 0 & \text{otherwise.} \end{cases}$$

To find $\mathbb{P}(1/4 < X < 3/4)$, use the marginal pdf $f_X(x)$:

$$\begin{aligned} \mathbb{P}(1/4 < X < 3/4) &= \int_{x=1/4}^{3/4} f_X(x)\, dx \\ &= \int_{x=1/4}^{3/4} \frac{6}{7}\left(2x^2 + x\right)\, dx \\ &= \left[\frac{6}{7}\left(\frac{2x^3}{3} + \frac{x^2}{2}\right)\right]_{x=1/4}^{3/4} \\ &= \left[\frac{1}{7}\left(4x^3 + 3x^2\right)\right]_{x=1/4}^{3/4} \\ &= \frac{1}{7}\left(\frac{27}{16} + \frac{27}{16} - \frac{1}{16} - \frac{3}{16}\right) \\ &= \frac{25}{56} \approx 0.4464 \end{aligned}$$

To find $\mathbb{P}(1/2 < Y)$, use the marginal pdf $f_Y(y)$:

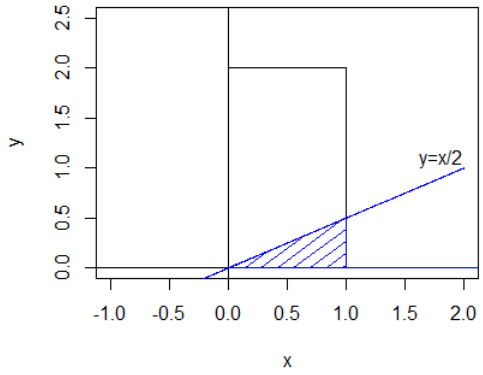
$$\begin{aligned} \mathbb{P}\left(1/2 < Y\right) &= \int_{y=1/2}^{\infty} f_Y(y)\, dy \\ &= \int_{y=1/2}^{2} \frac{2}{7} + \frac{3y}{14}\, dy \\ &= \left[\frac{2y}{7} + \frac{3y^2}{28}\right]_{y=1/2}^{2} \\ &= \frac{4}{7} + \frac{3}{7} - \frac{1}{7} - \frac{3}{112} \\ &= \frac{93}{112} \approx 0.8304 \end{aligned}$$

If X and Y were independent then $f_X(x) f_Y(y) = f_{X,Y}(x,y)$ for all $(x,y) \in \mathbb{R}^2$, therefore to show that X and Y are not independent it suffices to find a single pair (x,y) such that the product of the marginals differs from the joint density. One such example is $(1/2, 1/3)$, indeed,

$$\begin{aligned} f_X\left(\frac{1}{2}\right) &= \frac{6}{7}\left(2\left(\frac{1}{2}\right)^2 + \left(\frac{1}{2}\right)\right) = \frac{6}{7} \\ f_Y\left(\frac{1}{3}\right) &= \frac{2}{7} + \frac{3}{14}\left(\frac{1}{3}\right) = \frac{5}{14} \\ f_{X,Y}\left(\frac{1}{2}, \frac{1}{3}\right) &= \frac{6}{7}\left(\left(\frac{1}{2}\right)^2 + \frac{1}{2}\left(\frac{1}{2}\right)\left(\frac{1}{3}\right)\right) = \frac{2}{7}. \end{aligned}$$

Now $f_X\left(\frac{1}{2}\right) f_Y\left(\frac{1}{3}\right) = \frac{15}{49}$ which is different from $f_{X,Y}\left(\frac{1}{2}, \frac{1}{3}\right) = \frac{2}{7}$ therefore we can conclude that X and Y are not independent.

(d) To compute $\mathbb{P}(X > 2Y)$, integrate the joint density $f_{X,Y}(x, y)$ over the region in which the density is non-zero and $x > 2y$.



Integrate over shaded region:

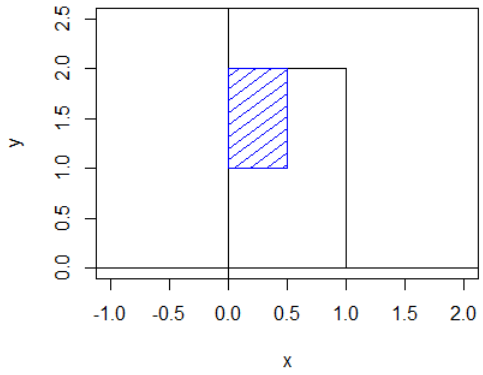
$$\begin{aligned}
\mathbb{P}(X > 2Y) &= \int_{x=0}^{1} \int_{y=0}^{x/2} f_{X,Y}(x, y)\, dy\, dx \\
&= \int_{x=0}^{1} \int_{y=0}^{x/2} \frac{6}{7}\left(x^2 + \frac{xy}{2}\right) dy\, dx \\
&= \int_{x=0}^{1} \left[\frac{6}{7}\left(x^2 y + \frac{xy^2}{4}\right)\right]_{y=0}^{x/2} dx \\
&= \int_{x=0}^{1} \frac{6}{7}\left(\frac{x^3}{2} + \frac{x^3}{16}\right) dx \\
&= \int_{x=0}^{1} \frac{27x^3}{56}\, dx \\
&= \left[\frac{27x^4}{224}\right]_{x=0}^{1} \\
&= \frac{27}{224} \approx 0.1205
\end{aligned}$$

(e) $\mathbb{P}(Y > 1 \mid X < 1/2) = \dfrac{\mathbb{P}(Y > 1 \text{ and } X < 1/2)}{\mathbb{P}(X < 1/2)}$

To compute the denominator, make use of the marginal density $f_X(x)$:

$$\begin{aligned}
\mathbb{P}(X < 1/2) &= \int_{x=0}^{1/2} f_X(x)\, dx \\
&= \int_{x=0}^{1/2} \frac{6}{7}\left(2x^2 + x\right) dx \\
&= \left[\frac{6}{7}\left(\frac{2x^3}{3} + \frac{x^2}{2}\right)\right]_{x=0}^{1/2} \\
&= \frac{6}{7}\left(\frac{1}{12} + \frac{1}{8}\right) = \frac{5}{28}
\end{aligned}$$

To compute the joint probability $\mathbb{P}(Y > 1 \text{ and } X < 1/2)$, need to integrate the joint density $f_{X,Y}(x, y)$ over the region in which the density is non-zero and $y > 1$ and $x < 1/2$.



Integrate over shaded region:

$$\begin{aligned}
\mathbb{P}(Y > 1 \text{ and } X < 1/2) &= \int_{x=0}^{1/2} \int_{y=1}^{2} f_{X,Y}(x,y)\, dy\, dx \\
&= \int_{x=0}^{1/2} \int_{y=1}^{2} \frac{6}{7}\left(x^2 + \frac{xy}{2}\right) dy\, dx \\
&= \int_{x=0}^{1/2} \left[\frac{6}{7}\left(x^2 y + \frac{xy^2}{4}\right)\right]_{y=1}^{2} dx \\
&= \int_{x=0}^{1/2} \frac{6}{7}\left(2x^2 + x - x^2 - \frac{x}{4}\right) dx \\
&= \int_{x=0}^{1/2} \frac{6}{7}\left(x^2 + \frac{3x}{4}\right) dx \\
&= \left[\frac{6}{7}\left(\frac{x^3}{3} + \frac{3x^2}{8}\right)\right]_{x=0}^{1/2} \\
&= \frac{6}{7}\left(\frac{1}{24} + \frac{3}{32}\right) = \frac{13}{112}
\end{aligned}$$

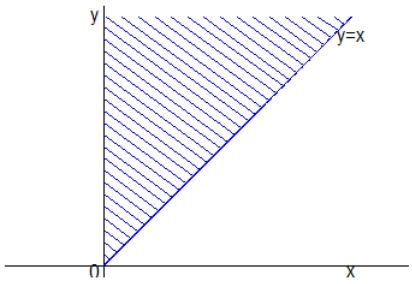
So required conditional probability is

$$\mathbb{P}(Y > 1 \mid X < 1/2) = \frac{\mathbb{P}(Y > 1 \text{ and } X < 1/2)}{\mathbb{P}(X < 1/2)} = \frac{13/112}{5/28} = \frac{13}{20} = 0.65$$

2. Joint probability density function of X and Y is

$$f_{X,Y}(x,y) = \begin{cases} ae^{-y/2} & 0 \le x \le y < \infty, \\ 0 & \text{otherwise.} \end{cases}$$

(a) Region of non-zero density:



Since $f_{X,Y}(x,y)$ is a density, the value of a must be such that the integral of $f_{X,Y}(x,y)$ over the shaded region is equal to 1. So

$$\begin{aligned}
1 = \iint_{\mathbb{R}^2} f_{X,Y}(x,y)\,dx\,dy &= \int_{y=0}^{\infty}\int_{x=0}^{y} ae^{-y/2}\,dx\,dy \\
&= \int_{y=0}^{\infty}\Big[axe^{-y/2}\Big]_{x=0}^{y} dy \\
&= \int_{y=0}^{\infty} aye^{-y/2}\,dy \\
&= \Big[-2aye^{-y/2}\Big]_{y=0}^{\infty} + \int_{y=0}^{\infty} 2ae^{-y/2}\,dy \text{ (integrating by parts)} \\
&= 0 + \Big[-4ae^{-y/2}\Big]_{y=0}^{\infty} \\
&= 4a \\
\Rightarrow \qquad a &= 1/4
\end{aligned}$$

(b) From the sketch above, can see that $\mathrm{Im}X = [0,\infty)$, so $f_X(x) = 0$ for $x < 0$. For any specific x value with $x \geq 0$, can see from the sketch above that the corresponding range of y values in the region of non-zero density is $y \geq x$. For $x \geq 0$, to find $f_X(x)$, integrate out y:

$$\begin{aligned}
f_X(x) = \int_{y=-\infty}^{\infty} f_{X,Y}(x,y)\,dy &= \int_{y=x}^{\infty} \frac{1}{4}e^{-y/2}\,dy \qquad \text{(recall that } a = 1/4\text{)} \\
&= \left[\frac{-e^{-y/2}}{2}\right]_{y=x}^{\infty} = \frac{1}{2}\,e^{-x/2}
\end{aligned}$$

That is,

$$f_X(x) = \begin{cases} \frac{1}{2}e^{-x/2} & x \geq 0, \\ 0 & \text{otherwise.} \end{cases}$$

Notice that $X \sim \mathrm{Exp}(\frac{1}{2})$.

Similarly, we can see from the shaded region that $\mathrm{Im}Y = [0,\infty)$, so $f_Y(y) = 0$ for $y < 0$. For $y \geq 0$, we see from the shaded region that the corresponding range of x values in the region of non-zero density is $0 \leq x \leq y$. For $y \geq 0$, to find $f_Y(y)$, integrate out x:

$$\begin{aligned}
f_Y(y) = \int_{x=-\infty}^{\infty} f_{X,Y}(x,y)\,dx &= \int_{x=0}^{y} \frac{1}{4}e^{-y/2}\,dx \\
&= \left[\frac{1}{4}e^{-y/2}x\right]_{x=0}^{y} = \frac{1}{4}ye^{-y/2}
\end{aligned}$$

That is,

$$f_Y(y) = \begin{cases} \frac{1}{4}ye^{-y/2} & y \geq 0, \\ 0 & \text{otherwise.} \end{cases}$$

Notice that $Y \sim \Gamma\left(2,\ \frac{1}{2}\right)$.

The variables X and Y are **not** independent, since within the shaded region,

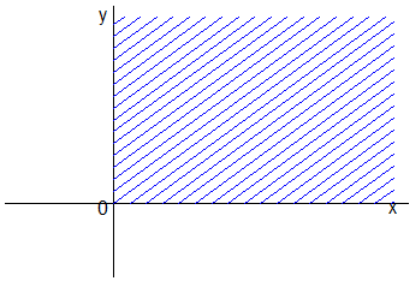
$$\begin{aligned} f_X(x)f_Y(y) &= \left(\frac{1}{2}e^{-x/2}\right)\left(\frac{1}{4}ye^{-y/2}\right) = \frac{ye^{-(x+y)/2}}{8} \\ &\neq f_{X,Y}(x,y). \end{aligned}$$

(Alternatively, observe that the shaded region of non-zero density is not a rectangle, so we can find a specific point outside the shaded area where $f_{X,Y}(x,y) = 0$ but $f_X(x) \neq 0$ and $f_Y(y) \neq 0$, eg at $(x,y) = (2,1)$.)

3. Joint probability density function of X and Y is

$$f_{X,Y}(x,y) = \begin{cases} x\exp(-x(y+1)) & x \geq 0,\ y \geq 0, \\ 0 & \text{otherwise.} \end{cases}$$

Region of non-zero density:



(a) From sketch, see that $\text{Im}X = [0,\infty)$, so $f_X(x) = 0$ for $x < 0$, and for $x \geq 0$,

$$\begin{aligned} f_X(x) = \int_{y=-\infty}^{\infty} f_{X,Y}(x,y)\,dy &= \int_{y=0}^{\infty} xe^{-x(y+1)}\,dy \\ &= xe^{-x}\int_{y=0}^{\infty} e^{-xy}\,dy \\ &= xe^{-x}\left[\frac{-e^{-xy}}{x}\right]_{y=0}^{\infty} = xe^{-x}\left(\frac{1}{x}\right) = e^{-x} \end{aligned}$$

That is,

$$f_X(x) = \begin{cases} e^{-x} & x \geq 0, \\ 0 & \text{otherwise.} \end{cases}$$

Notice that $X \sim \text{Exp}(1)$.

From sketch of non-zero density region, see that $\text{Im}Y = [0, \infty)$, so $f_Y(y) = 0$ for $y < 0$, and for $y \geq 0$,

$$\begin{aligned} f_Y(y) &= \int_{x=-\infty}^{\infty} f_{X,Y}(x,y)\,dx \\ &= \int_{x=0}^{\infty} xe^{-x(y+1)}\,dx \\ &= \left[-\frac{xe^{-x(y+1)}}{y+1}\right]_{x=0}^{\infty} + \int_{x=0}^{\infty} \frac{e^{-x(y+1)}}{y+1}\,dx \qquad \text{(integrating by parts)} \\ &= 0 + \left[\frac{-e^{-x(y+1)}}{(y+1)^2}\right]_{x=0}^{\infty} = \frac{1}{(y+1)^2} \end{aligned}$$

That is,

$$f_Y(y) = \begin{cases} \frac{1}{(y+1)^2} & y \geq 0, \\ 0 & \text{otherwise.} \end{cases}$$

To compute $\mathbb{P}(X > 1)$, use marginal density of X:

$$\begin{aligned} \mathbb{P}(X > 1) = \int_{x=1}^{\infty} f_X(x)\,dx &= \int_{x=1}^{\infty} e^{-x}\,dx \\ &= \left[-e^{-x}\right]_{x=1}^{\infty} = e^{-1} \approx 0.3679. \end{aligned}$$

To compute $\mathbb{P}(1 \leq Y \leq 2)$, use marginal density of Y:

$$\begin{aligned} \mathbb{P}(1 \leq Y \leq 2) = \int_{y=1}^{2} f_Y(y)\,dy &= \int_{y=1}^{2} \frac{1}{(y+1)^2}\,dy \\ &= \left[\frac{-1}{(y+1)}\right]_{y=1}^{2} = -\frac{1}{3} + \frac{1}{2} = \frac{1}{6} \approx 0.1667. \end{aligned}$$

(b) For $y \geq 0$, the conditional density of X given that $Y = y$ is

$$\begin{aligned} f_{X|Y}(x \mid y) = \frac{f_{X,Y}(x,y)}{f_Y(y)} &= \frac{xe^{-x(y+1)}}{1/(y+1)^2} \text{ for } x \geq 0 \\ &= (y+1)^2 xe^{-x(y+1)} \text{ for } x \geq 0. \end{aligned}$$

Also notice that $f_{X|Y}(x \mid y) = 0$ for $x < 0$.
So for $y \geq 0$,

$$f_{X|Y}(x \mid y) = \begin{cases} (y+1)^2 xe^{-x(y+1)} & x \geq 0, \\ 0 & \text{otherwise.} \end{cases}$$

Notice that this is the $\Gamma(2,\ y+1)$ density. So the conditional distribution of X given that $Y = y$ (where $y \geq 0$) is the $\Gamma(2,\ y+1)$ distribution.
For $x \geq 0$, the conditional density of Y given that $X = x$ is

$$\begin{aligned} f_{Y|X}(y \mid x) = \frac{f_{X,Y}(x,y)}{f_X(x)} &= \frac{xe^{-x(y+1)}}{e^{-x}} \text{ for } y \geq 0 \\ &= xe^{-xy} \text{ for } x \geq 0. \end{aligned}$$

Also notice that $f_{Y|X}(y \mid x) = 0$ for $y < 0$.
So for $x \geq 0$,

$$f_{Y|X}(y \mid x) = \begin{cases} xe^{-xy} & y \geq 0, \\ 0 & \text{otherwise.} \end{cases}$$

Notice that this is the Exp(x) density. So the conditional distribution of Y given that $X = x$ (where $x \geq 0$) is the Exp(x) distribution.
To determine $\mathbb{P}(X > 1 \mid Y = 1)$, use the conditional density $f_{X|Y}(x \mid 1)$:

$$\begin{aligned}
\mathbb{P}(X > 1 \mid Y = 1) &= \int_{x=1}^{\infty} f_{X|Y}(x \mid 1)\, dx \\
&= \int_{x=1}^{\infty} 4xe^{-2x}\, dx \\
&= \left[-2xe^{-2x}\right]_{x=1}^{\infty} + \int_{x=1}^{\infty} 2e^{-2x}\, dx \qquad \text{(integrating by parts)} \\
&= 2e^{-2} + \left[-e^{-2x}\right]_{x=1}^{\infty} \\
&= 2e^{-2} + e^{-2} = 3e^{-2} \approx 0.4060.
\end{aligned}$$

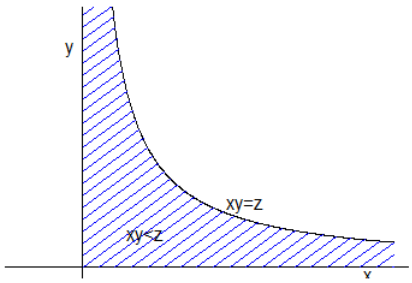
To determine $\mathbb{P}(1 \leq Y \leq 3 \mid X = 2)$, use the conditional density $f_{Y|X}(y \mid 2)$:

$$\begin{aligned}
\mathbb{P}(1 \leq Y \leq 3 \mid X = 2) &= \int_{y=1}^{3} f_{Y|X}(y \mid 2)\, dy \\
&= \int_{y=1}^{3} 2e^{-2y}\, dy \\
&= \left[-e^{-2y}\right]_{y=1}^{3} = e^{-2} - e^{-6} \approx 0.1329.
\end{aligned}$$

(c) With $Z = XY$, to find the density function $f_Z(z)$ of Z, the first step is to find the cumulative distribution function $F_Z(z)$. Since $\mathrm{Im}X = [0, \infty)$ and $\mathrm{Im}Y = [0, \infty)$, we have $\mathrm{Im}Z = [0, \infty)$, and so $F_Z(x) = 0$ for $z < 0$. For $z \geq 0$,

$$F_Z(z) = \mathbb{P}(Z \leq z) = \mathbb{P}(XY \leq x)$$

For a specific (fixed) values of z, need to integrate over shaded region below:



$$\begin{aligned} F_Z(z) = \mathbb{P}(Z \le z) &= \mathbb{P}(XY \le x) \\ &= \int_{x=0}^{\infty} \int_{y=0}^{z/x} f_{X,Y}(x,y)\, dy\, dx \\ &= \int_{x=0}^{\infty} \int_{y=0}^{z/x} x e^{-x(y+1)}\, dy\, dx = \int_{x=0}^{\infty} \int_{y=0}^{z/x} x e^{-x} e^{-xy}\, dy\, dx \\ &= \int_{x=0}^{\infty} \left[x e^{-x} \left(\frac{e^{-xy}}{-x} \right) \right]_{y=0}^{z/x} dx = \int_{x=0}^{\infty} \left[-e^{-x} e^{-xy} \right]_{y=0}^{z/x} dx \\ &= \int_{x=0}^{\infty} -e^{-x} e^{-z} + e^{-x}\ dx \\ &= \left[e^{-x} e^{-z} - e^{-x} \right]_{x=0}^{\infty} = 1 - e^{-z} \end{aligned}$$

That is,

$$F_Z(z) = \begin{cases} 0 & z < 0, \\ 1 - e^{-z} & z \ge 0. \end{cases}$$

To find the density function $f_Z(z)$, differentiate:

$$f_Z(z) = \begin{cases} 0 & z < 0, \\ e^{-z} & z \ge 0. \end{cases}$$

Notice that $Z \sim \text{Exp}(1)$.

4. Random variables X, Y have joint density

$$f_{X,Y}(x,y) = \begin{cases} a e^{-\lambda x - \mu y} & x \ge 0,\ y \ge 0, \\ 0 & \text{otherwise.} \end{cases}$$

(a) Densities must integrate to 1, so (noting that $\lambda > 0$ and $\mu > 0$)

$$\begin{aligned} 1 = \iint_{\mathbb{R}^2} f_{X,Y}(x,y)\, dy\, dx &= \int_{x=0}^{\infty} \int_{y=0}^{\infty} a e^{-\lambda x - \mu y}\, dy\, dx \\ &= \int_{x=0}^{\infty} \left[-\frac{a e^{-\lambda x - \mu y}}{\mu} \right]_{y=0}^{\infty} dx = \int_{x=0}^{\infty} \frac{a e^{-\lambda x}}{\mu}\, dx \\ &= \left[-\frac{a e^{-\lambda x}}{\mu \lambda} \right]_{x=0}^{\infty} = \frac{a}{\mu \lambda} \\ \Rightarrow \qquad a &= \mu \lambda. \end{aligned}$$

(b) To find marginal density $f_X(x)$, first notice that $\text{Im}X = [0, \infty)$, so $f_X(x) = 0$ for $x < 0$. For $x \ge 0$, integrate out y:

$$f_X(x) = \int_{y=-\infty}^{\infty} f_{X,Y}(x,y)\, dy = \int_{y=0}^{\infty} \mu \lambda e^{-\lambda x - \mu y}\, dy = \left[-\lambda e^{-\lambda x - \mu y} \right]_{y=0}^{\infty} = \lambda e^{-\lambda x}$$

That is,

$$f_X(x) = \begin{cases} 0 & x < 0, \\ \lambda e^{-\lambda x} & x \ge 0. \end{cases}$$

Similarly, $\text{Im}Y = [0, \infty)$, so $f_Y(y) = 0$ for $y < 0$. For $y \ge 0$, integrate out x:

$$f_Y(y) = \int_{x=-\infty}^{\infty} f_{X,Y}(x,y)\, dx = \int_{x=0}^{\infty} \mu \lambda e^{-\lambda x - \mu y}\, dx = \left[-\mu e^{-\lambda x - \mu y} \right]_{x=0}^{\infty} = \mu e^{-\mu y}$$

That is,

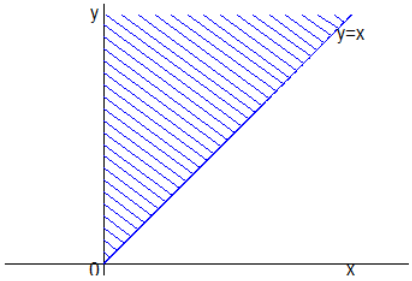
$$f_Y(y) \quad = \quad \begin{cases} 0 & y < 0, \\ \mu e^{-\mu y} & y \geq 0. \end{cases}$$

Notice that $X \sim \text{Exp}(\lambda)$ and $Y \sim \text{Exp}(\mu)$.
For $x \geq 0$, $y \geq 0$ we have

$$f_X(x) f_Y(y) = \left(\lambda e^{-\lambda x}\right)\left(\mu e^{-\mu y}\right) = \mu\lambda e^{-\lambda x - \mu y} = f_{X,Y}(x, y).$$

Also, for $x < 0$ or $y < 0$ we have $f_X(x) f_Y(y) = 0 = f_{X,Y}(x, y)$.
So X and Y are independent.

(c) To determine $\mathbb{P}(X < Y)$ integrate over the shaded region:



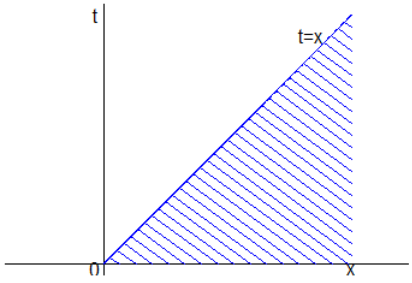
So

$$\begin{aligned} \mathbb{P}(X < Y) \quad &= \quad \int_{y=0}^{\infty} \int_{x=0}^{y} \mu\lambda e^{-\lambda x - \mu y}\, dx\, dy \\ &= \quad \int_{y=0}^{\infty} \left[-\mu e^{-\lambda x - \mu y}\right]_{x=0}^{y} dy = \int_{y=0}^{\infty} -\mu e^{-(\lambda+\mu)y} + \mu e^{-\mu y}\, dy \\ &= \quad \left[\frac{\mu e^{-(\lambda+\mu)y}}{\lambda+\mu} - e^{-\mu y}\right]_{y=0}^{\infty} = 1 - \frac{\mu}{\lambda+\mu} = \frac{\lambda}{\lambda+\mu} \end{aligned}$$

5. (a) If X is a continuous random variable with $\text{Im}X \subseteq [0, \infty)$, and density $f_X(x)$, then

$$\int_{t=0}^{\infty} \mathbb{P}(X > t)\, dt \quad = \quad \int_{t=0}^{\infty} \left(\int_{x=t}^{\infty} f_X(x)\, dx \right) dt$$

Region of integration is shaded area below:



So changing the order of integration,

$$\begin{aligned}\int_{t=0}^{\infty} \mathbb{P}(X > t)\, dt &= \int_{t=0}^{\infty}\int_{x=t}^{\infty} f_X(x)\, dx\, dt \\ &= \int_{x=0}^{\infty}\int_{t=0}^{x} f_X(x)\, dt\, dx \\ &= \int_{x=0}^{\infty} [t f_X(x)]_{t=0}^{x}\, dx \\ &= \int_{x=0}^{\infty} x f_X(x)\, dx \\ &= \mathbb{E}(X),\end{aligned}$$

as required.

(b) With $X \sim \text{Exp}(\lambda)$ we have $\text{Im}X = [0, \infty)$ and $\mathbb{P}(X > t) = 1 - F_X(t) = 1 - (1 - e^{-\lambda t}) = e^{-\lambda t}$, so using the result from part (a),

$$\mathbb{E}(X) = \int_{t=0}^{\infty} \mathbb{P}(X > t)\, dt = \int_{t=0}^{\infty} e^{-\lambda t}\, dt = \left[-\frac{e^{-\lambda t}}{\lambda}\right]_{t=0}^{\infty} = \frac{1}{\lambda} \qquad \text{(noting that } \lambda > 0\text{)}$$

Exam Style Question

Let X and Y be jointly continuous random variables with density function

$$f_{X,Y}(x,y) = \begin{cases} 7x^5 & \text{if } 0 < y < x < 1 \\ 0 & \text{otherwise.} \end{cases}$$

(a) To show that $f_{X,Y}(x,y)$ is a valid joint density function we need to establish that $f_{X,Y}(x,y) \geq 0$ for all $x, y \in \mathbb{R}$ and that $\iint_{\mathbb{R}^2} f_{X,Y}\, \mathrm{d}x\, \mathrm{d}y = 1$.

Note that $f_{X,Y})(x,y) = 0$ unless $0 < y < x < 1$ and otherwise $x > 0$ so $x^5 > 0$ thus $f_{X,Y}(x,y) \geq 0$ for all values of x and y.

Thus it remains to show $\iint_{\mathbb{R}^2} f_{X,Y}\, \mathrm{d}x\, \mathrm{d}y = 1$.

$$\begin{aligned}\iint_{\mathbb{R}^2} f_{X,Y}\, \mathrm{d}x\, \mathrm{d}y &= \int_{x=0}^{1}\int_{y=0}^{x} 7x^5\, \mathrm{d}y\, \mathrm{d}x \\ &= \int_{x=0}^{1} \left[7x^5 y\right]_{y=0}^{x} \mathrm{d}x \\ &= \int_{x=0}^{1} 7x^6\, \mathrm{d}x \\ &= \left[x^7\right]_{x=0}^{x=1} = 1.\end{aligned}$$

So $f_{X,Y}(x,y)$ is a valid joint density function. (2 marks)

(b) For the X marginal:

$$f_X(x) = \int_{y=0}^{x} 7x^5\, \mathrm{d}y = 7x^6, \quad \text{for } 0 < x < 1$$

and $f_X(x) = 0$ otherwise. (2 marks)

For the Y marginal:

$$f_Y(y) = \int_{x=y}^{1} 7x^5\, \mathrm{d}x = \frac{7}{6}(1 - y^6) \quad \text{for } 0 < y < 1$$

and $f_Y(y) = 0$ otherwise. (2 marks)

(c) X and Y are not independent since $f_{X,Y}(x,y) \neq f_X(x)f_Y(y)$ we can establish this by checking the point $(1/4, 1/2)$. Indeed

$$\begin{aligned} f_{X,Y}(1/4,1/2) &= 0 \\ f_X(1/4) &= \frac{7}{4^6} \neq 0 \\ f_Y(1/2) &= \frac{7}{6}\left(1 - \frac{1}{2^6}\right) \neq 0. \end{aligned}$$

From this it is clear that $f_{X,Y}(1/4,1/2) = 0$ however $f_X(1/4)f_Y(1/2) \neq 0$. (1 mark)

(d)

$$\mathbb{E}[X] = \int_{-\infty}^{\infty} x f_X(x)\,\mathrm{d}x = \int_0^1 x(7x^6)\,\mathrm{d}x = \frac{7}{8}.$$

(1 mark)

$$\mathbb{E}[Y] = \int_{-\infty}^{\infty} f_Y(y)\,\mathrm{d}y = \frac{7}{6}\int_0^1 y(1-y^6)\,\mathrm{d}y = \frac{7}{6}\left[\frac{y^2}{2} - \frac{y^8}{8}\right]_{y=0}^{1} = \frac{7}{16}.$$

(1 mark)

(e) Let $Z = Y/X$. Find the cumulative distribution function F_Z of Z. (6 marks)

Note that $0 < Y < X < 1$ so $0 < Y/X < 1$ therefore $F_Z(z) = 0$ for $z \leq 0$ and $F_Z(z) = 1$ for $z \geq 1$. (2 marks)

For $z \in (0,1)$,

$$\begin{aligned} F_Z(z) &= \mathbb{P}(Z \leq z) \\ &= \mathbb{P}\left(\frac{Y}{X} \leq z\right) \\ &= \mathbb{P}(Y \leq zX) \\ &= \int_{x=0}^{1}\int_{y=0}^{zx} 7x^5\,\mathrm{d}y\,\mathrm{d}x \\ &= \int_{x=0}^{1}\left[7x^6 y\right]_{y=0}^{zx}\mathrm{d}x \\ &= \int_{x=0}^{1} 7zx^7\,\mathrm{d}x \\ &= \left[zx^7\right]_{x=0}^{1} \\ &= z. \end{aligned}$$

(4 marks)

So we have

$$F_Z(z) = \begin{cases} 0 & z \leq 0, \\ z & 0 < z < 1, \\ 1 & z \geq 1. \end{cases}$$